

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_11apr25_ajust.parquet

model_type: LSTM
roll_window: 2
timesteps_orig: 60
timesteps: 30
batch_size: 256
min_units: 32
max_units: 256
units_step: 32
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 150
hyperband_iterations: 3
forecast_steps: 20

LSTM Multi-Step Training Report

Report Generated: 2025-05-08 11:48:34

Training Start: 2025-05-08_10-52-10

Training End: 2025-05-08_11-20-06

Training Duration: 27.93 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: elu

Optimizer: rmsprop

Dropout Rate: 0.20

MAE: 1.7786

MSE: 4.5405

RMSE: 2.1308

R2 Score: 0.2368

Multi-Step Forecast:

Forecast Steps: 20

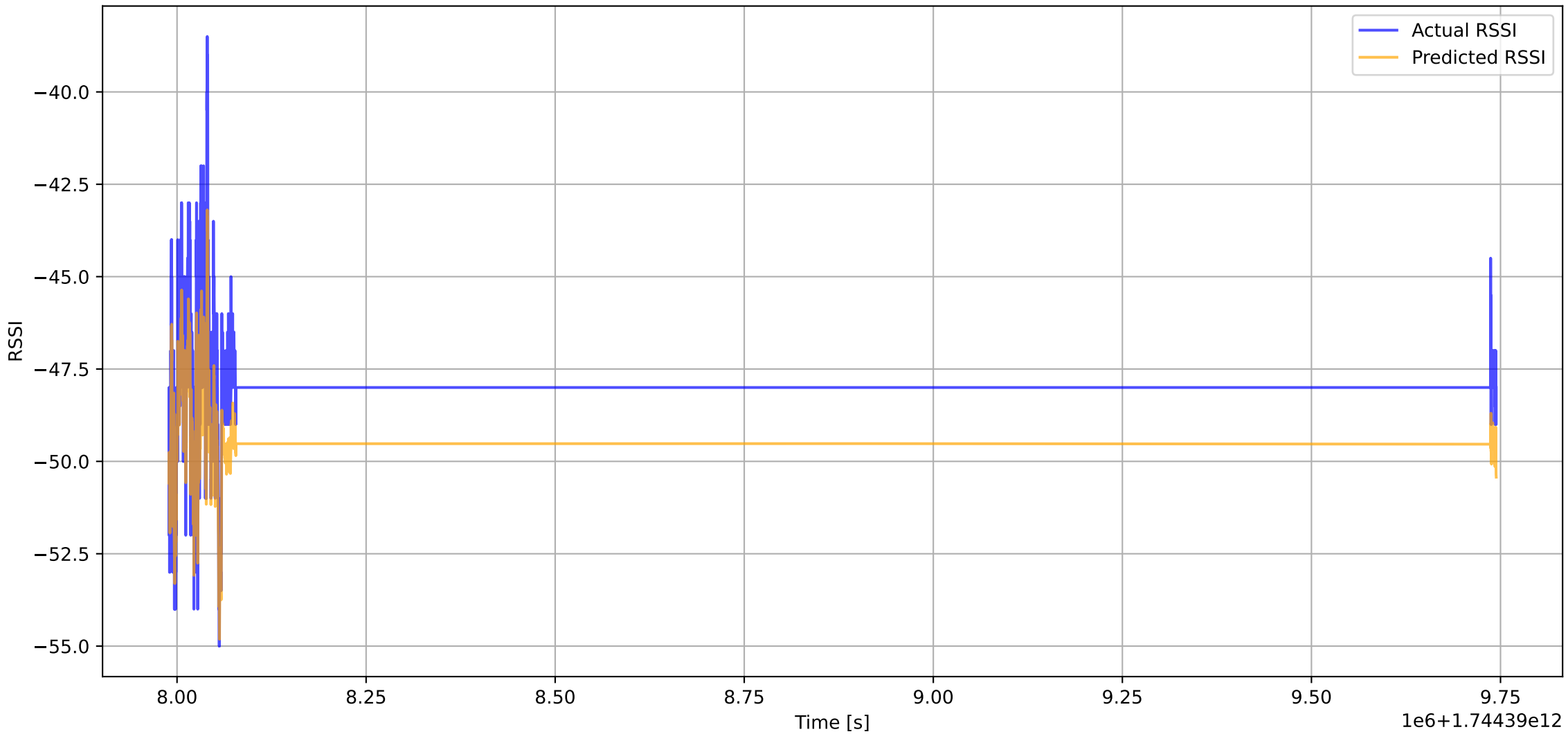
MAE: 0.0537

MSE: 0.0038

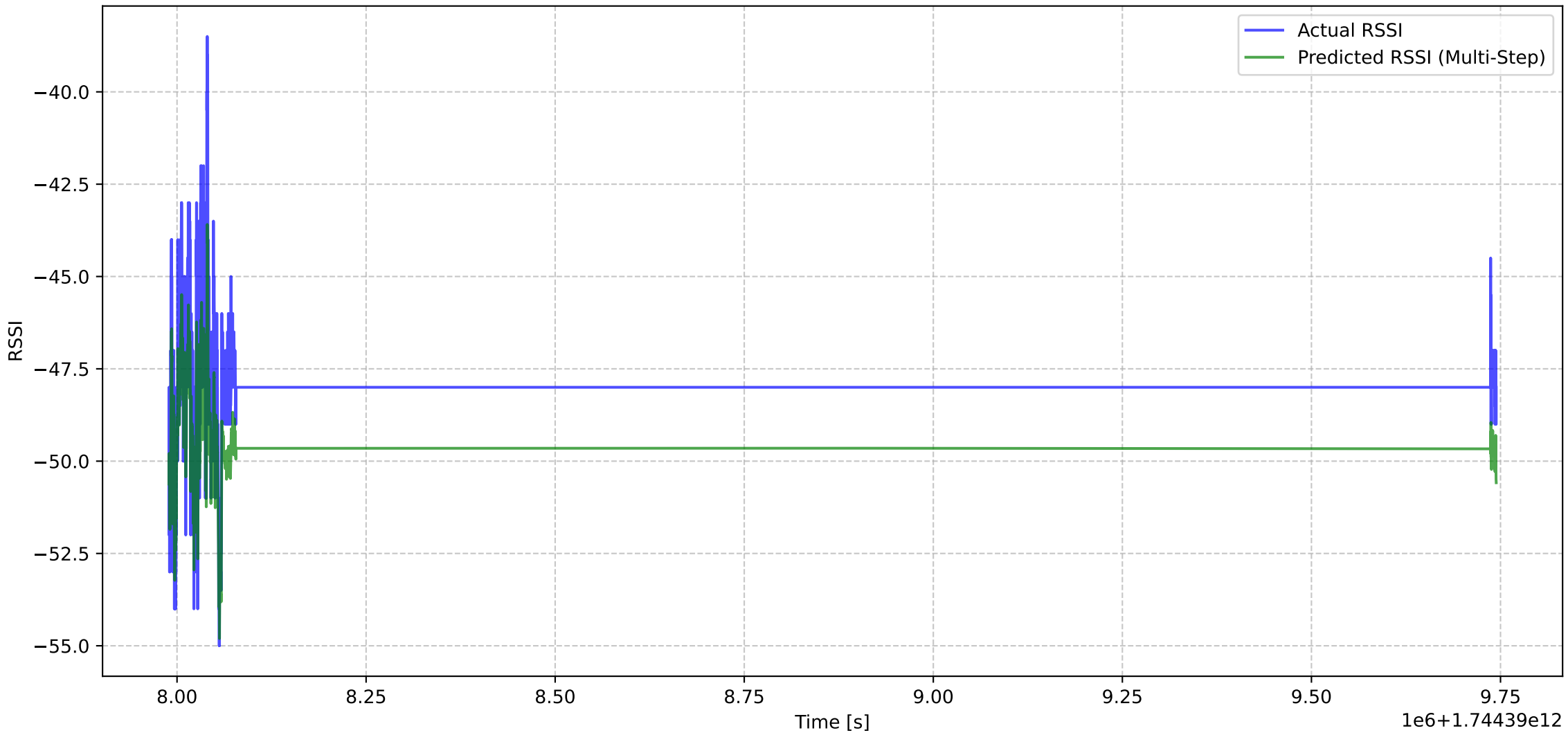
RMSE: 0.0616

R2 Score: 0.4845

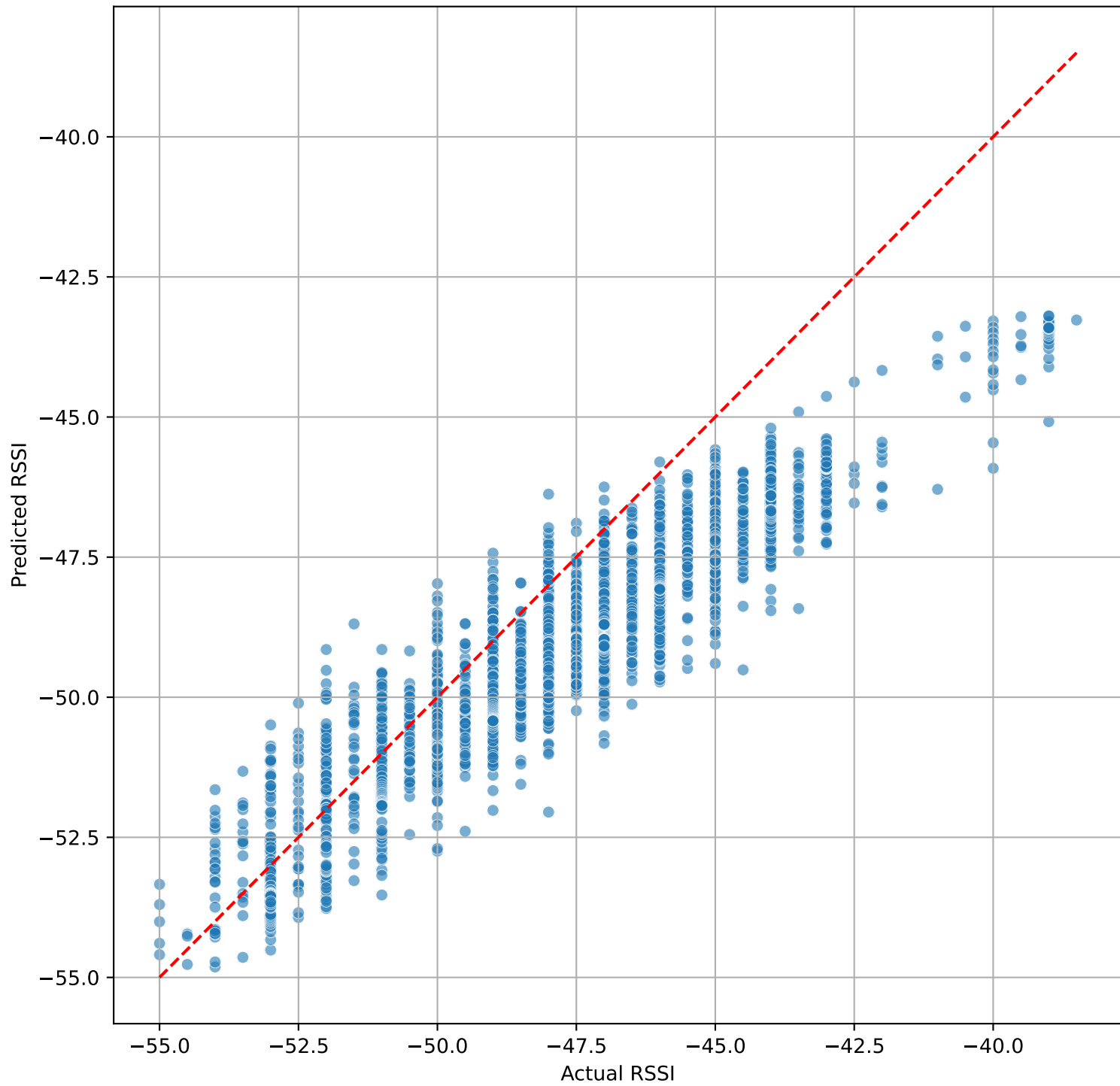
Actual vs Predicted RSSI (First Step)



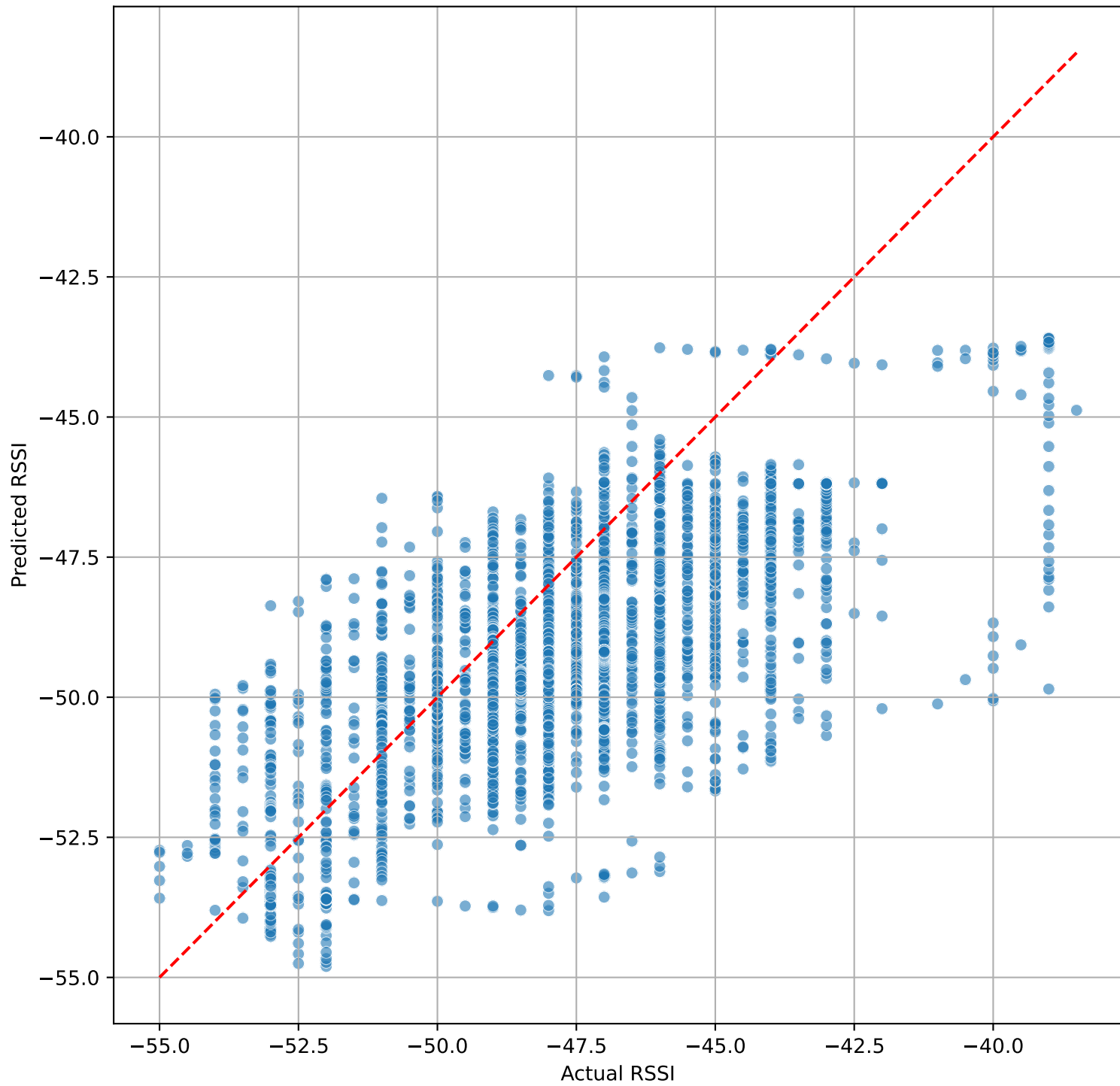
Actual vs Predicted RSSI (Multi-Step (step 20))



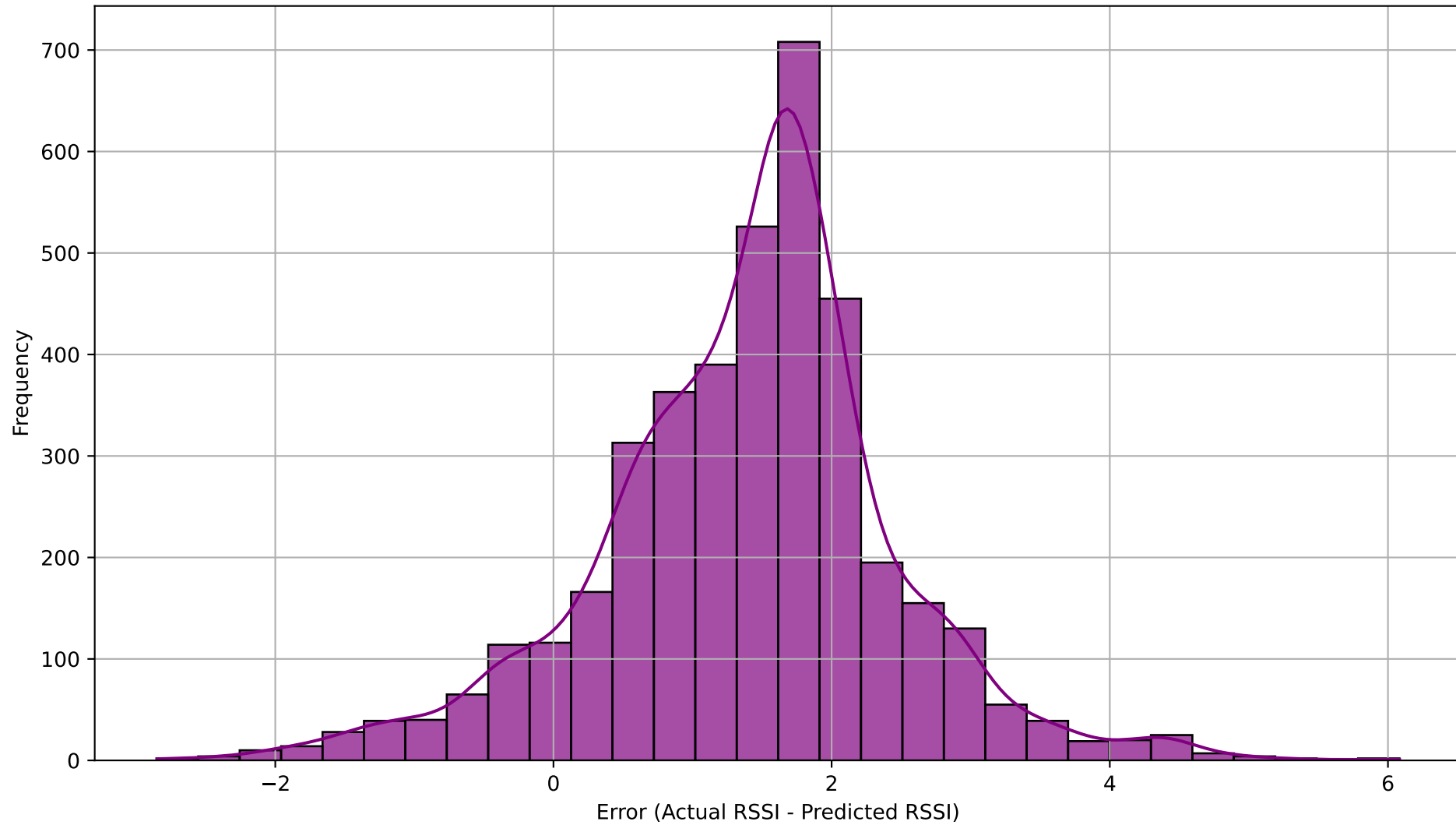
Actual vs Predicted RSSI (First Step)



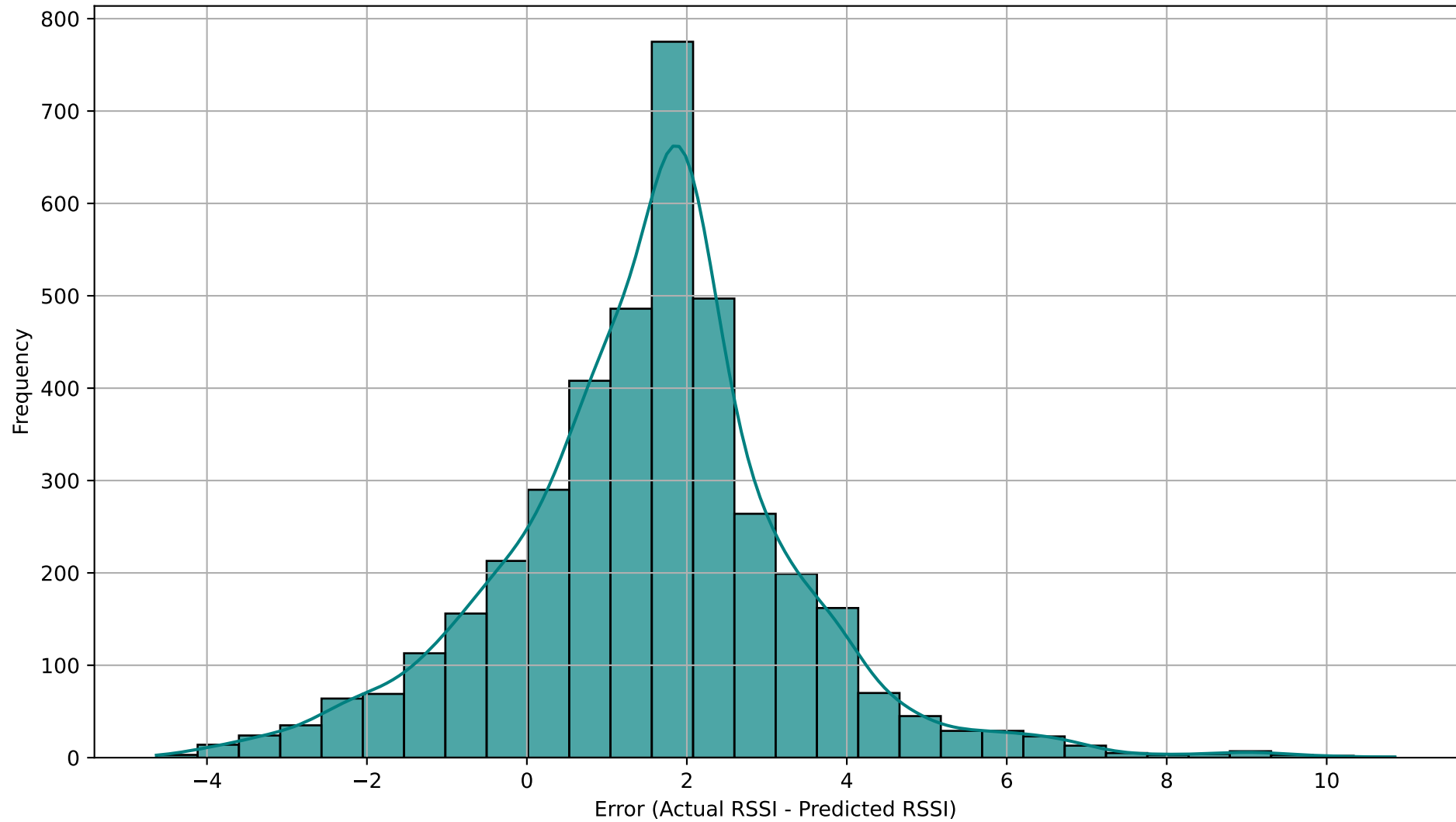
Actual vs Predicted RSSI (Last Step - step 20)



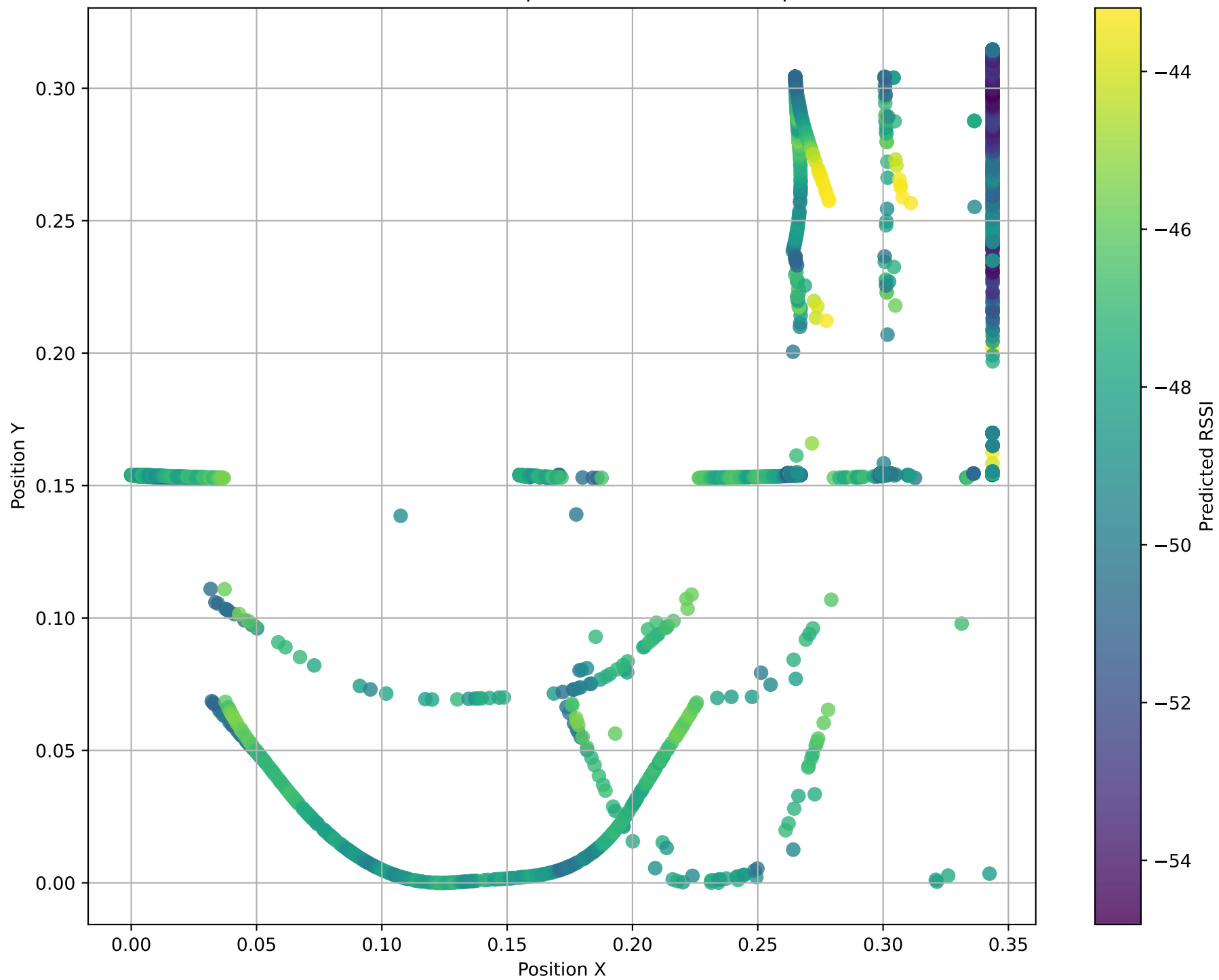
Distribution of Prediction Errors (First Step)



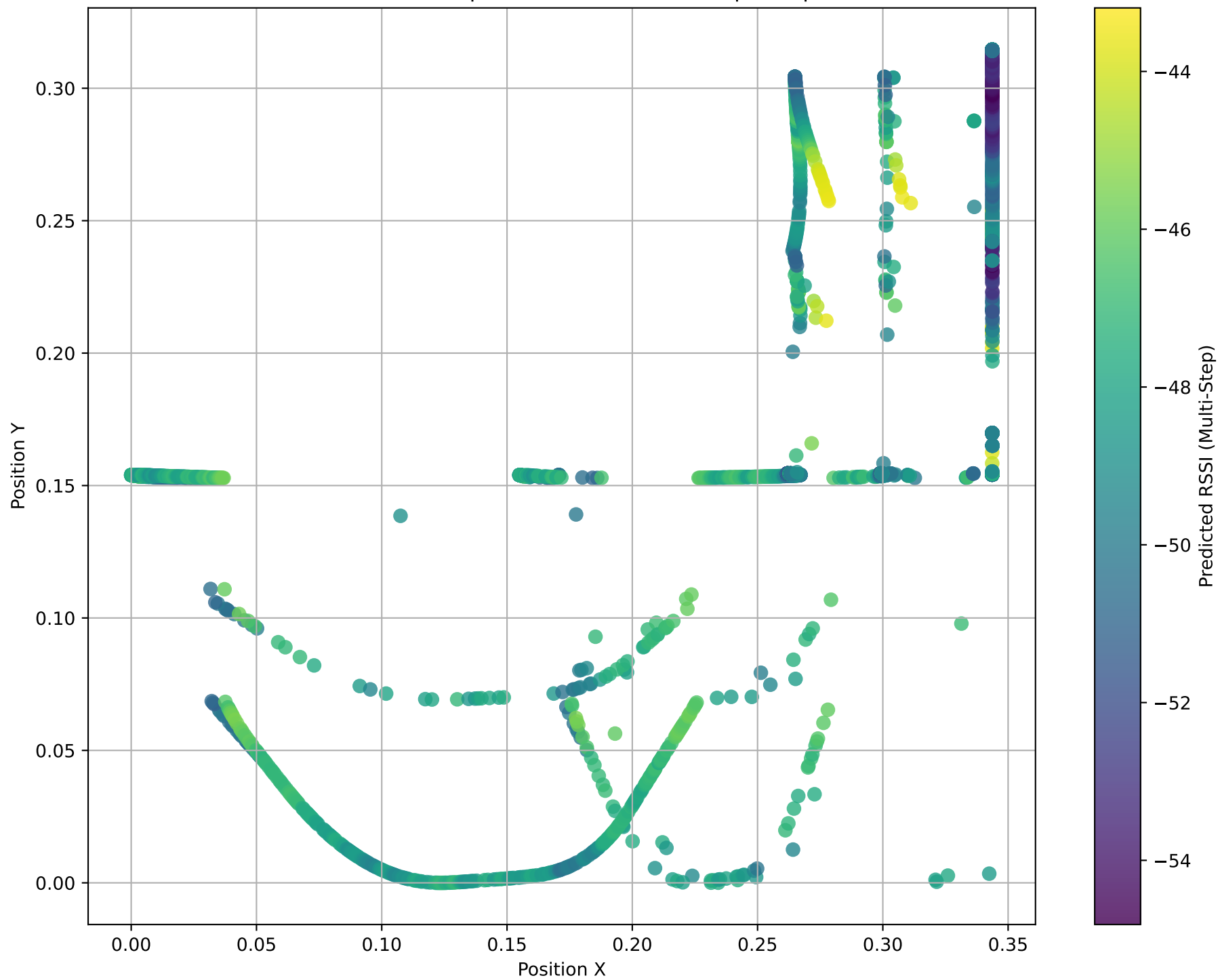
Distribution of Prediction Errors (Last Step - step 20)



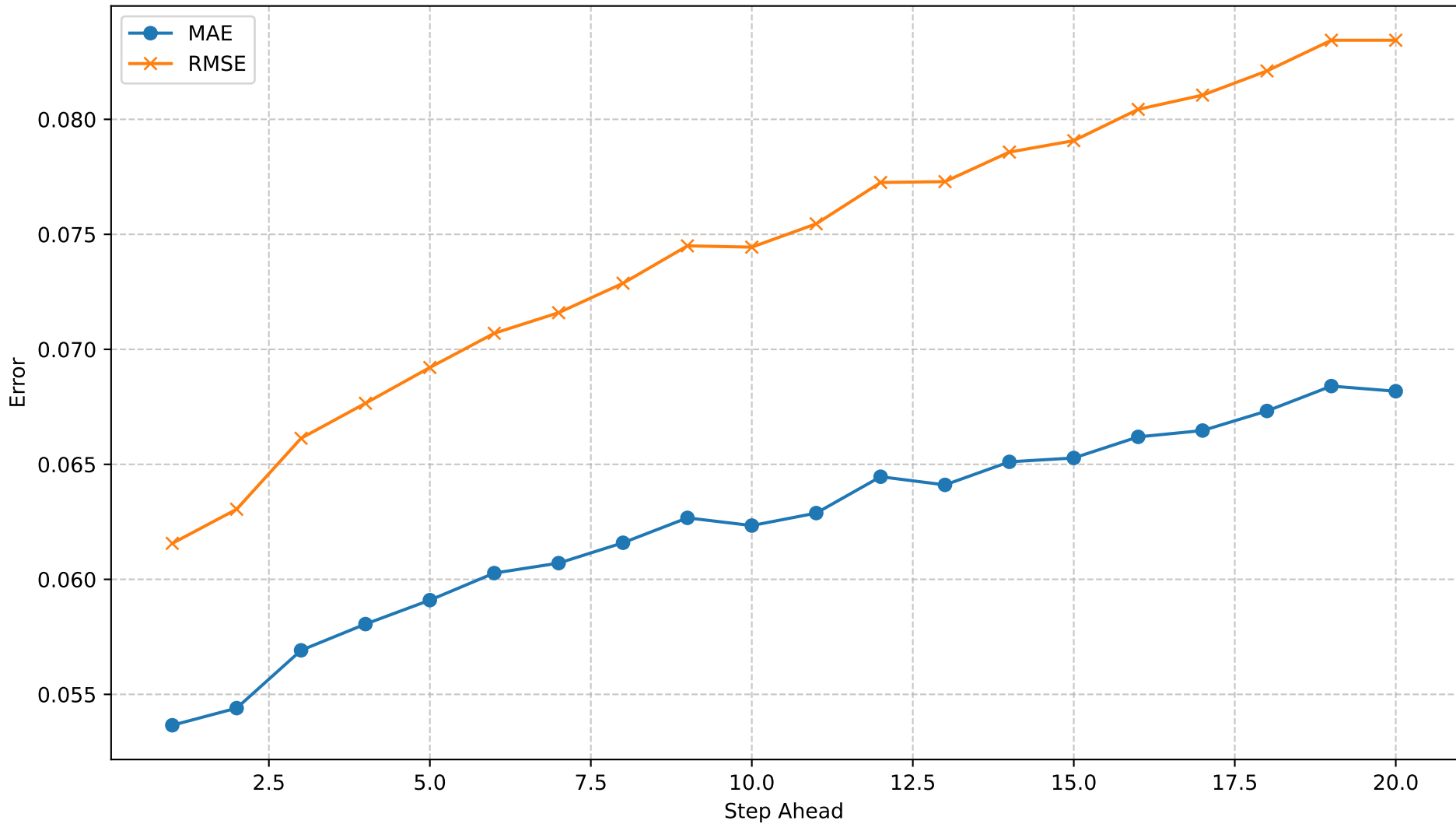
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 20)



Multi-Step Forecast Error by Step



Resumo Automático

Dataset: rssi_odom_dataset_11apr25_ajust.parquet

Modelo: LSTM

Forecast steps: 20

MAE / RMSE (dB)

- 1º passo : 0.05 / 0.06
- Mediana : 0.06 / 0.07
- 20º passo : 0.07 / 0.08

Crescimento do erro: 27.1% → Moderado (erro cresce 20-50 %)

3 piores passos (MAE): 19 (0.07), 20 (0.07), 18 (0.07)

Recomendação:

Use as previsões até a metade do horizonte; após isso a incerteza cresce consideravelmente.